

EE25C01 – Basic Electrical & Electronics Engineering

Last-Minute Exam Survival Guide

1. MOST IMPORTANT TOPICS (Study in this order)

Tier 1 – Do these first, no matter what (covers ~60% of marks):

- Ohm's Law, KCL, KVL, series-parallel circuits
- RMS, average, peak, form factor, power factor (AC basics)
- PN junction diode + rectifiers (half-wave, full-wave)
- Number system conversions, logic gates, Boolean algebra
- EMF equation of DC generator & transformer

Tier 2 – Do these next:

- Thevenin's / Norton's / Superposition theorem

- Single-phase RLC series circuit, impedance, phasor diagram
- BJT basics, CE configuration
- Half adder, full adder, flip-flops (SR, D, JK)
- Induction motor principle, slip

Tier 3 – If time remains:

- Transformer losses & efficiency
- 8051 microcontroller architecture, microprocessor vs microcontroller
- K-map, universal gates, De Morgan's theorems

If you only have 3 hours, **Tier 1 + Tier 2 alone can get you a comfortable pass**, since they repeat across 1-mark, 3-mark, and 12-mark questions.

2. TOP 20 EXPECTED 1-MARK QUESTIONS

1. **State Ohm's Law.** → $V = IR$ (at constant temperature).
2. **State KCL.** → Sum of currents entering a node = sum leaving it.
3. **State KVL.** → Sum of voltage drops in a closed loop = 0.

4. **Define RMS value.** → DC equivalent value that produces the same heating effect as the AC over one full cycle.
5. **Define power factor.** → $\cos \theta = \text{Real Power} / \text{Apparent Power}$.
6. **What is a PN junction diode?** → Device formed by joining P-type and N-type semiconductor; conducts in one direction only.
7. **Use of Zener diode?** → Voltage regulation, operates in reverse breakdown region.
8. **What is rectification?** → Process of converting AC to DC.
9. **Define ripple factor.** → Ratio of AC component (RMS) to DC component in rectifier output.
10. **What is a transformer?** → Static device that transfers electrical energy between circuits via mutual induction, frequency unchanged.
11. **Principle of DC generator?** → Faraday's Law of Electromagnetic Induction.
12. **Define slip in induction motor.** → $\text{Slip} = (N_s - N_r) / N_s$, where N_s = synchronous speed, N_r = rotor speed.
13. **Convert binary 1011 to decimal.** → 11
14. **State De Morgan's first theorem.** → $(A+B)' = A' \cdot B'$
15. **What is a half adder?** → Combinational circuit adding two bits, giving Sum and Carry outputs.
16. **Name the universal gates.** → NAND and NOR

17. **What is a flip-flop?** → Basic 1-bit memory element with two stable states.
18. **One-line difference: microprocessor vs microcontroller.** → Microprocessor needs external RAM/ROM/I/O; microcontroller has CPU+memory+I/O on one chip.
19. **Function of accumulator in 8051?** → Holds operand/result for arithmetic & logic operations.
20. **Define form factor.** → Form Factor = RMS value / Average value (=1.11 for pure sine wave).
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3. TOP 20 EXPECTED 3-MARK QUESTIONS

1. **State Superposition theorem.** In a linear circuit with multiple sources, total current/voltage in any branch = sum of currents/voltages produced by each source acting alone (other sources replaced by their internal resistance, voltage sources shorted, current sources opened).
2. **Voltage Divider Rule.** In a series circuit, voltage across any resistor $R_1 = V \times R_1 / (R_1 + R_2 + \dots)$. Used to find voltage drop without computing total current first.
3. **Series vs Parallel circuits.**

- Series: same current, voltages add, total $R = R_1 + R_2 + \dots$
- Parallel: same voltage, currents add, $1/R = 1/R_1 + 1/R_2 + \dots$

4. **Thevenin's theorem (short note).** Any linear two-terminal network can be replaced by a single voltage source (V_{th}) in series with a single resistance (R_{th}). V_{th} = open-circuit voltage at terminals; R_{th} = resistance seen from terminals with sources removed.

5. **Peak factor & Form factor for sine wave.** Peak factor = $V_m/V_{rms} = 1.414$. Form factor = $V_{rms}/V_{avg} = 1.11$. These tell us how "peaky" a waveform is compared to its average/RMS.

6. **Power triangle.** Apparent Power S (VA) = VI , Active Power P (Watt) = $VI \cos\theta$, Reactive Power Q (VAR) = $VI \sin\theta$. Related by $S^2 = P^2 + Q^2$.

7. **Forward vs Reverse bias of PN diode.**

- Forward bias: P connected to +, N to -; barrier reduces, current flows easily.
- Reverse bias: P to -, N to +; barrier increases, only tiny leakage current flows.

8. **Half-wave rectifier.** Single diode conducts only during positive half-cycle of AC input, blocking the

negative half. Output is pulsating DC for half the cycle only. $V_{dc} = V_m/\pi$.

9. **Full-wave bridge rectifier.** Uses 4 diodes; conducts during both halves of AC cycle (different diode pairs for each half), giving smoother DC. $V_{dc} = 2V_m/\pi$.

10. **BJT vs FET (any 3 points).**

- BJT is current-controlled, FET is voltage-controlled.
- BJT has 3 terminals (Base, Emitter, Collector); FET has (Gate, Source, Drain).
- FET has very high input impedance; BJT has lower input impedance.

11. **EMF equation of DC generator.** $E = (P\Phi NZ)/(60A)$, where P =poles, Φ =flux/pole, N =speed (rpm), Z =total armature conductors, A =number of parallel paths.

12. **EMF equation of transformer.** $E = 4.44 f \Phi_m N$, where f =frequency, Φ_m =max flux, N =number of turns. Shows EMF is directly proportional to frequency and flux.

13. **Losses in a transformer.**

- Copper loss (I^2R , in windings, varies with load)
- Hysteresis loss (in core, due to repeated magnetization)

- Eddy current loss (circulating currents in core, reduced by laminating core)

14. Principle of 3-phase induction motor. The three-phase supply to stator windings produces a rotating magnetic field (RMF). This RMF induces current in the rotor (by induction), which creates its own field, causing the rotor to follow the RMF, but always slightly slower (hence "induction" and "slip").

15. Decimal to Binary conversion (example: 25). $25 \div 2$ repeatedly: $25 = 11001_2$. Show division-remainder method, read remainders bottom to top.

16. De Morgan's theorems.

- $(A+B)' = A' \cdot B'$
- $(A \cdot B)' = A' + B'$

Used to convert OR/AND logic into NAND/NOR forms (useful for universal gate implementation).

17. Full adder. Adds three bits (A, B, Carry-in) giving Sum and Carry-out. $\text{Sum} = A \oplus B \oplus C_{in}$, $\text{Carry-out} = AB + BC_{in} + AC_{in}$. Built using two half adders + OR gate.

18. Combinational vs Sequential circuits.

- Combinational: output depends only on present inputs (e.g., adders, gates), no

memory.

- Sequential: output depends on present input AND past state (e.g., flip-flops, counters), has memory.

19. **SR Flip-flop.** Has Set (S) and Reset (R) inputs.

$S=1, R=0 \rightarrow Q=1$ (set). $S=0, R=1 \rightarrow Q=0$ (reset).

$S=0, R=0 \rightarrow$ no change (hold). $S=1, R=1 \rightarrow$ invalid/forbidden state.

20. **8051 memory organization (brief).** Has internal RAM (128/256 bytes) for data, internal ROM (4KB) for program code, plus 4 I/O ports (P0–P3), each 8-bit, used for input/output operations.

4. TOP 10 EXPECTED 12-MARK QUESTIONS (Detailed Answers)

Q1. State and explain Thevenin's Theorem with an example.

Statement: Any linear bilateral network containing voltage/current sources and resistances, viewed from two terminals, can be replaced by an equivalent circuit consisting of a single voltage source (V_{th}) in series with a single resistance (R_{th}).

Steps to solve using Thevenin's theorem:

1. Remove the load resistor (R_L) from the circuit – the open terminals are A-B.
2. Find V_{th} = open-circuit voltage across A-B (using normal circuit analysis, voltage divider, etc., with R_L removed).
3. Find R_{th} = equivalent resistance across A-B with all independent voltage sources shorted and current sources opened.
4. Redraw simple equivalent circuit: V_{th} in series with R_{th} , then reconnect R_L .
5. Load current $I_L = V_{th} / (R_{th} + R_L)$.

Example: For a circuit with a 10V source, series resistor 2Ω , and load $R_L=3\Omega$ connected after a parallel combination – calculate V_{th} by removing R_L and finding open-circuit voltage, then R_{th} by deactivating the source and finding resistance seen from open terminals, then I_L using the formula above.

Advantage: Very useful when you need to study the effect of changing only the load resistance, without re-solving the whole circuit each time.

Q2. Analyze a Single-Phase Series RLC circuit (Impedance, Phasor Diagram, Power Factor).

Circuit: R, L, C connected in series with AC supply V (frequency f).

Reactances:

- $X_L = 2\pi fL$ (inductive reactance, opposes increase in current, causes voltage to lead current by 90°)
- $X_C = 1/(2\pi fC)$ (capacitive reactance, causes voltage to lag current by 90°)

Impedance: $Z = R + j(X_L - X_C)$, magnitude $|Z| = \sqrt{R^2 + (X_L - X_C)^2}$

Current: $I = V/Z$

Phase angle: $\theta = \tan^{-1}((X_L - X_C)/R)$

- If $X_L > X_C$: circuit is inductive, current lags voltage (lagging p.f.)
- If $X_L < X_C$: circuit is capacitive, current leads voltage (leading p.f.)
- If $X_L = X_C$: resonance, circuit is purely resistive, p.f.=1

Power factor: $\cos\theta = R/Z$

Phasor diagram (description): Draw I as reference (horizontal). V_R is in phase with I . V_L is drawn 90° ahead of I (vertical, upward). V_C is drawn 90° behind I (vertical, downward). Resultant V is the vector sum of V_R , V_L , V_C — drawn at angle θ from I .

Power: Active power $P = VI \cos\theta$; Reactive power $Q = VI \sin\theta$; Apparent power $S = VI$.

Q3. Explain Construction, Principle, and EMF Equation of a DC Generator.

Construction: Main parts are the yoke (outer frame), poles & pole shoes (produce magnetic field), field winding (creates flux when current flows), armature core (rotating part holding conductors), armature winding (where EMF is induced), commutator (converts induced AC to DC by reversing connections every half rotation), and brushes (carry current from commutator to external circuit).

Principle: Based on Faraday's Law — whenever a conductor moves through a magnetic field, an EMF is induced in it. The armature is rotated mechanically (by a prime mover) inside the magnetic field of the poles, so EMF is continuously induced in the armature conductors; the commutator+brush arrangement converts this into unidirectional (DC) output.

EMF equation derivation: For one revolution, flux cut by one conductor = $P\Phi$ (P poles, Φ flux/pole). Time for one revolution = $60/N$ seconds (N in rpm). So EMF per conductor = $P\Phi N/60$. For Z total conductors arranged in A parallel paths, total EMF:

$$E = \frac{P\Phi ZN}{60A}$$

Types (by excitation): Separately excited (field winding powered by external source), Self-excited — further divided into Shunt (field winding in parallel

with armature), Series (field winding in series with armature), and Compound (combination of both).

Q4. Explain Construction, Principle, EMF Equation, Losses and Efficiency of a Transformer.

Construction: Two windings (primary, secondary) wound on a common laminated iron core. Core provides a low-resistance magnetic path; laminations reduce eddy current loss.

Principle: Mutual induction — AC supply to the primary creates a changing flux in the core, which links with the secondary winding and induces an EMF in it (frequency stays same, voltage changes per turns ratio).

EMF equation: $E = 4.44 f \Phi_m N$ (f =frequency, Φ_m =max flux, N =turns). Voltage ratio: $E_2/E_1 = N_2/N_1 = K$ (turns ratio).

Losses:

- Copper loss (I^2R in primary & secondary windings; varies with load — "variable loss")
- Iron/Core loss = Hysteresis loss (energy lost magnetizing/demagnetizing core repeatedly) + Eddy current loss (circulating currents induced in

core, reduced by laminating); these are constant regardless of load.

Efficiency: $\eta = \text{Output Power} / (\text{Output Power} + \text{Losses}) \times 100$, generally highest near 75-100% load.

Voltage regulation: % Regulation = $((\text{No-load voltage} - \text{Full-load voltage}) / \text{Full-load voltage}) \times 100$; tells how much the secondary voltage drops from no-load to full-load.

Q5. Explain the Construction and Working Principle of a Three-Phase Induction Motor.

Construction: Stator (stationary part with 3-phase winding fed by 3-phase AC supply) and Rotor (rotating part) – two types: Squirrel Cage Rotor (simple bars short-circuited by end rings, rugged, low cost) and Slip Ring (Wound) Rotor (windings connected to external resistance via slip rings, used for speed control/high starting torque).

Working Principle: When 3-phase AC supply is given to the stator, it produces a Rotating Magnetic Field (RMF) that rotates at synchronous speed $N_s = 120f/P$. This rotating field cuts the stationary rotor conductors, inducing EMF in them (by induction – hence the name). This induced EMF drives current in the short-circuited rotor, which creates its own

magnetic field. By Lenz's law, the rotor tries to catch up with the RMF, so it rotates — but it can never reach exactly N_s (if it did, there would be no relative motion, no induced EMF, no torque). The difference is called slip: $s = (N_s - N_r)/N_s$.

Torque-slip relationship: Torque is zero at $s=0$ (synchronous speed, no relative motion) and at $s=1$ (standstill, torque is the starting torque). Torque increases with slip up to a maximum (pull-out torque), then decreases — giving the characteristic torque-slip curve.

Advantages: Self-starting (squirrel cage type), simple, rugged, low maintenance, widely used in industry.

Q6. Explain the Construction, V-I Characteristics and Applications of a PN Junction Diode.

Construction: Formed by joining a P-type semiconductor (excess holes) and N-type semiconductor (excess electrons). At the junction, a "depletion region" forms naturally due to diffusion, creating a potential barrier ($\sim 0.7V$ for silicon).

Forward bias: P connected to positive terminal, N to negative. This reduces the barrier, and once supply

voltage exceeds barrier potential, current flows easily and increases sharply with small voltage increase.

Reverse bias: P connected to negative, N to positive. This widens the depletion region, and only a tiny reverse saturation/leakage current flows — until "breakdown voltage" is reached, where current increases sharply (this is what Zener diodes exploit).

V-I characteristic curve (description): In the forward region, current stays near zero until $\sim 0.7\text{V}$ (knee voltage), then rises steeply. In the reverse region, current is almost zero (μA range) until breakdown voltage, where it rises sharply.

Applications: Rectification (AC to DC), switching circuits, clipping/clamping circuits, signal demodulation. Zener diodes specifically are used for voltage regulation.

Q7. Explain Half-Wave and Full-Wave Rectifiers with Circuit Diagrams and Derive V_{dc} .

Half-Wave Rectifier:

- Circuit: single diode in series with load, fed by AC supply via transformer.
- Working: diode conducts only during the positive half-cycle (when anode is positive relative to

cathode); blocks the negative half-cycle completely.

- Output: pulsating DC, present only half the time.
- $V_{dc} = V_m/\pi$ (where V_m = peak AC voltage)
- Efficiency $\approx 40.6\%$, Ripple factor ≈ 1.21 (high ripple, poor quality output)

Full-Wave Rectifier (Bridge type):

- Circuit: four diodes arranged in a bridge configuration with AC supply across one diagonal and load across the other.
- Working: for each half-cycle, a different diagonal pair of diodes conducts, so current always flows through the load in the same direction — output is present for the full cycle.
- $V_{dc} = 2V_m/\pi$
- Efficiency $\approx 81.2\%$, Ripple factor ≈ 0.48 (much smoother DC output)

Comparison: Full-wave is preferred in practice because it gives higher DC output, better efficiency, and lower ripple than half-wave, even though it needs 4 diodes instead of 1.

Q8. Explain the Construction, Working and CE Characteristics of a BJT (Bipolar Junction Transistor).

Construction: Three regions — Emitter (heavily doped, supplies carriers), Base (very thin, lightly doped), Collector (moderately doped, collects carriers). Two types: NPN and PNP, based on doping arrangement.

Working (NPN, in CE configuration): Emitter-Base junction is forward biased (injects electrons into the base), Collector-Base junction is reverse biased. Most electrons from the emitter diffuse through the thin base and are swept into the collector due to the reverse bias, forming collector current. A small base current controls a much larger collector current — this is the basis of amplification.

CE Characteristics:

- Input characteristic: I_B vs V_{BE} (similar to diode curve)
- Output characteristic: I_C vs V_{CE} for different I_B values — shows three regions:
 - **Cut-off region:** both junctions reverse biased, transistor acts as OFF switch ($I_C \approx 0$)
 - **Saturation region:** both junctions forward biased, transistor acts as ON switch (I_C max)
 - **Active region:** EB forward biased, CB reverse biased — used for amplification, $I_C \approx \beta \cdot I_B$

Applications: Amplifier (active region), Switch (cut-off/saturation regions), oscillators.

Q9. Explain Boolean Algebra Basics, Logic Gates, Universal Gates, and Design a Full Adder.

Basic gates and truth tables:

- AND: output=1 only if all inputs=1
- OR: output=1 if any input=1
- NOT: inverts input
- NAND = AND followed by NOT (universal gate)
- NOR = OR followed by NOT (universal gate)
- XOR: output=1 if inputs differ
- XNOR: output=1 if inputs are same

Why NAND/NOR are "universal": Any logic function (AND, OR, NOT) can be built using only NAND gates (or only NOR gates), which makes IC manufacturing simpler and cheaper — only one type of gate needed for any circuit.

Boolean laws (key ones): Commutative ($A+B=B+A$), Associative, Distributive ($A(B+C)=AB+AC$), Identity ($A+0=A$, $A \cdot 1=A$), De Morgan's (as above).

Full Adder design: Adds 3 bits — A, B, and Carry-in (C_{in}) — producing Sum and Carry-out (C_{out}).

- $\text{Sum} = A \oplus B \oplus C_{in}$
- $\text{Cout} = AB + BC_{in} + AC_{in}$

- **Truth table:** for all 8 combinations of A, B, Cin, list Sum and Cout (e.g., 0,0,0 → S=0,C=0; 1,1,1 → S=1,C=1, etc.)
 - **Implementation:** built from two Half Adders + one OR gate — first half adder adds A and B, second half adder adds that result with Cin, OR gate combines the two carry outputs.
 - **Use:** building block for multi-bit binary adders (used in ALUs/calculators).
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Q10. Explain the Architecture of 8051 Microcontroller with Block Diagram.

Overview: 8051 is an 8-bit microcontroller with CPU, memory, and I/O all on a single chip — making it self-contained for embedded applications.

Main blocks:

- **CPU (Central Processing Unit):** performs arithmetic/logic operations; includes the Accumulator (main register for operations) and B register.
- **Internal ROM (4KB):** stores the program code permanently.
- **Internal RAM (128 bytes, sometimes 256 in extended versions):** temporary data storage during program execution.

- **4 I/O Ports (P0, P1, P2, P3):** each 8-bit, used to interface with external devices (sensors, LEDs, switches etc.); P0 & P2 also used for external memory addressing when needed.
- **Two 16-bit Timers/Counters:** used for timing operations and counting external events.
- **Serial Port:** for serial communication (UART) with other devices.
- **Interrupt Control:** allows the CPU to respond immediately to external/internal events (5 interrupt sources in standard 8051) instead of constantly checking ("polling").
- **Oscillator/Clock circuit:** provides timing pulses to synchronize all internal operations.
- **PSW (Program Status Word):** flag register holding status bits like Carry, Zero, etc., used after arithmetic/logic operations.
- **Program Counter (PC) & Stack Pointer (SP):** PC holds address of next instruction to execute; SP manages the stack for subroutine calls.

Why important: This single-chip integration is exactly why microcontrollers (unlike microprocessors) don't need external RAM/ROM/I/O chips for basic applications — making them cheap, compact, and ideal for embedded systems like washing machines, remote controls, etc.

5. THREE-HOUR LAST-MINUTE STUDY PLAN

Hour 1 (0:00 – 1:00) – DC + AC Fundamentals

- 0:00–0:20 – Ohm's Law, KCL, KVL, series-parallel rules (Section 6 formulas)
- 0:20–0:40 – Thevenin's theorem steps + Q1 (12-mark) once
- 0:40–1:00 – RMS/average/form factor/power factor + Q2 (12-mark RLC) once

Hour 2 (1:00 – 2:00) – Semiconductor + Digital Electronics

- 1:00–1:20 – PN diode + rectifiers (Q6, Q7) – these repeat heavily in exams
- 1:20–1:40 – Logic gates, Boolean laws, half/full adder (Q9)
- 1:40–2:00 – Number system conversions + flip-flops (quick practice, 2-3 examples)

Hour 3 (2:00 – 3:00) – Electric Machines + Microcontrollers + Mop-up

- 2:00–2:20 – DC generator & transformer EMF equations (Q3, Q4)
- 2:20–2:35 – Induction motor principle & slip (Q5)
- 2:35–2:50 – 8051 basics (Q10) + microprocessor vs microcontroller

- 2:50–3:00 – Skim through all 20 one-mark answers once, fast

Tomorrow morning (30 min before exam): Use Section 7 only – don't open anything else.

6. KEY FORMULAS, DEFINITIONS & DIAGRAMS (Cheat Sheet)

DC Fundamentals

- Ohm's Law: $V = IR$
- Series R: $R_{\text{total}} = R_1 + R_2 + R_3 \dots$
- Parallel R: $1/R_{\text{total}} = 1/R_1 + 1/R_2 + \dots$
- Power: $P = VI = I^2R = V^2/R$

AC Fundamentals

- Instantaneous value: $v = V_m \sin(\omega t)$, $\omega = 2\pi f$
- RMS value (sine wave) = $V_m/\sqrt{2} = 0.707 V_m$
- Average value (sine wave, half cycle) = $2V_m/\pi = 0.637 V_m$
- Form factor = 1.11, Peak factor = 1.414
- Impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$
- $X_L = 2\pi fL$, $X_C = 1/(2\pi fC)$
- Power factor = $\cos\theta = R/Z$

Semiconductor Devices

- Half-wave $V_{dc} = V_m/\pi$
- Full-wave $V_{dc} = 2V_m/\pi$
- Diode forward knee voltage (Si) $\approx 0.7V$, (Ge) $\approx 0.3V$
- BJT: $I_C = \beta \cdot I_B$ (β = current gain, typically 50–200)

Electric Machines

- DC Generator EMF: $E = P\Phi ZN/(60A)$
- Transformer EMF: $E = 4.44f\Phi_m N$
- Transformer turns ratio: $E_2/E_1 = N_2/N_1 = I_1/I_2$
- Synchronous speed: $N_s = 120f/P$
- Slip: $s = (N_s - N_r)/N_s$

Digital Electronics

- Binary → Decimal: multiply each bit by $2^{(\text{position})}$, sum up
- Decimal → Binary: repeated division by 2, read remainders bottom-to-top
- De Morgan's: $(A+B)' = A'B'$, $(AB)' = A'+B'$
- Half Adder: $\text{Sum} = A \oplus B$, $\text{Carry} = AB$
- Full Adder: $\text{Sum} = A \oplus B \oplus C_{in}$, $C_{out} = AB + BC_{in} + AC_{in}$

Microcontrollers

- 8051: 8-bit CPU, 4KB ROM, 128B RAM, 4 ports (P0–P3), 2 timers, 1 serial port, 5 interrupts

Simple diagrams to sketch quickly if asked (just rough shapes are fine for marks):

- *PN junction*: box split into P (left, with – – – symbols for holes) and N (right, with + + + for electrons), thin depletion region in the middle.
 - *Half-wave rectifier*: AC source → diode (arrow+bar symbol) → load resistor, with output waveform showing only top halves of the sine wave appearing, bottom halves flat at zero.
 - *Full-wave bridge*: AC source connected to one diagonal of a diamond of 4 diodes, load connected to other diagonal; output waveform shows all humps pointing upward (no negative half).
 - *Transformer*: two coils wound on a common rectangular laminated core, primary on left fed by AC, secondary on right connected to load.
 - *Logic gate symbols*: AND = D-shaped, OR = curved-back shield shape, NOT = triangle with small circle at tip, NAND/NOR = AND/OR shape with small circle added at output.
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7. FINAL 30-MINUTE REVISION SHEET (Read this right before the exam, nothing else)

Must-remember formulas:

- $V=IR$ | $R_{\text{series}}=\Sigma R$ | $1/R_{\text{parallel}}=\Sigma(1/R)$
- $V_{\text{rms}}=0.707V_m$ | Form factor=1.11 | Peak factor=1.414
- $Z=\sqrt{R^2+(X_L-X_C)^2}$ | p.f.= $\cos\theta=R/Z$
- Half-wave $V_{\text{dc}}=V_m/\pi$ | Full-wave $V_{\text{dc}}=2V_m/\pi$
- DC Gen EMF: $E=P\Phi ZN/60A$
- Transformer EMF: $E=4.44f\Phi_m N$
- $N_s=120f/P$ | Slip $s=(N_s-N_r)/N_s$

Must-remember one-liners:

- KCL = currents in = currents out at a node
- KVL = sum of voltage drops in a loop = 0
- Thevenin's = single V_{th} + R_{th} replacing a network
- PN diode conducts only in forward bias
- Zener diode → voltage regulation (reverse breakdown)
- Full-wave rectifier > half-wave (less ripple, more efficient)
- BJT: small base current controls large collector current
- NAND, NOR = universal gates
- Half adder = 2 bits → Sum+Carry; Full adder = 3 bits → Sum+Carry
- Flip-flop = 1-bit memory; SR has a forbidden state ($S=1, R=1$)

- Induction motor never reaches synchronous speed (always has slip)
- Microcontroller = CPU+RAM+ROM+I/O on ONE chip; microprocessor needs external chips
- 8051: 4KB ROM, 128B RAM, 4 ports, 2 timers

Exam strategy reminder:

- Attempt ALL 1-mark and 3-mark questions first — they're pure recall, fast marks.
- For 12-mark questions, write whatever you remember in structured points (principle → diagram → working → formula → applications) — partial answers still get partial marks, so never leave a 12-mark question blank.
- If a derivation formula is forgotten, at least write the final formula and explain it in words — examiners often give marks for correct concept even without full derivation.

Good luck for tomorrow — you know more than you think once it's organized like this.